

The Spinal Cord and the Ascending and Descending Tracts



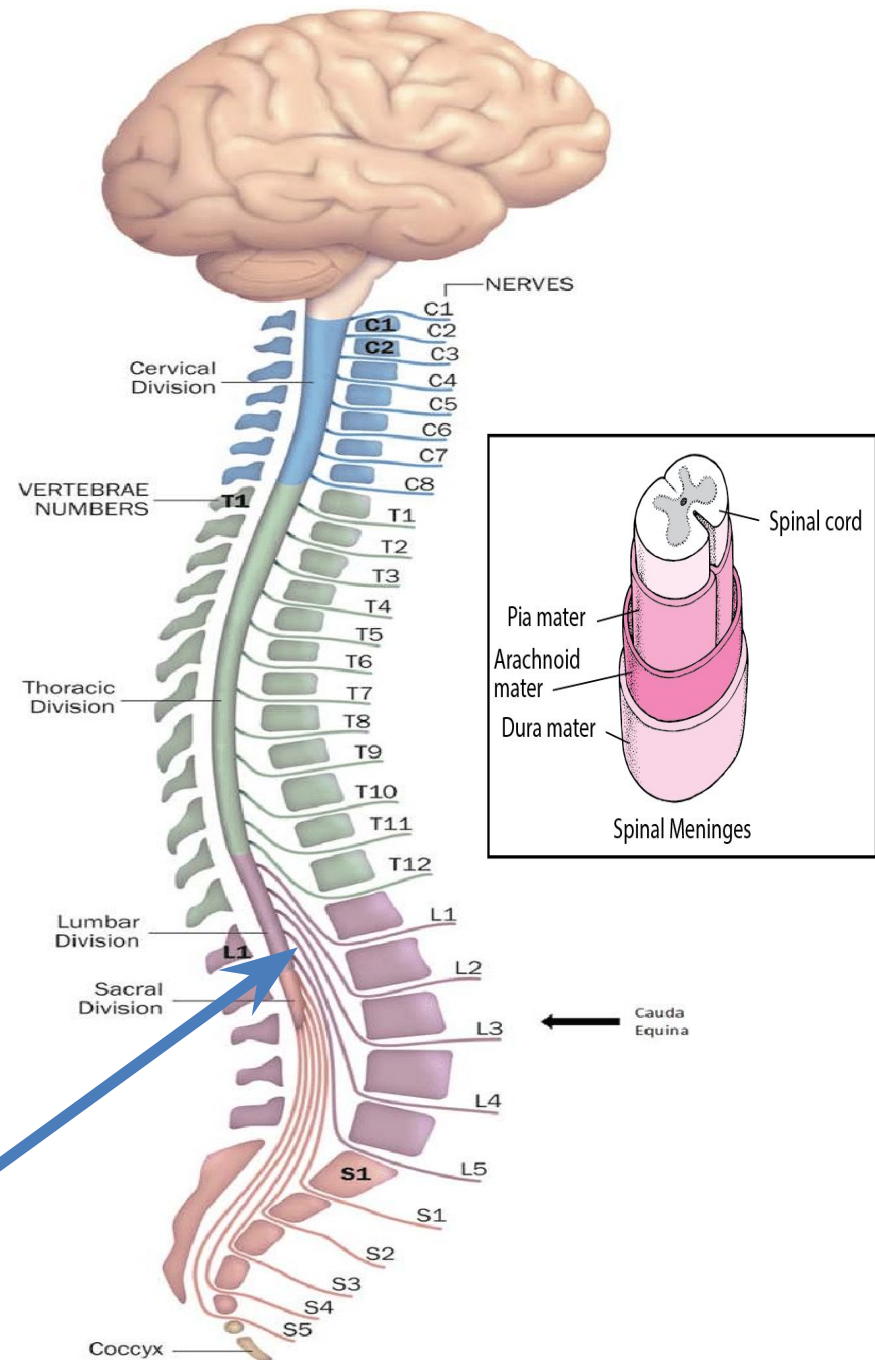
Learning objectives:

At the end of the lecture, the student should be able to:

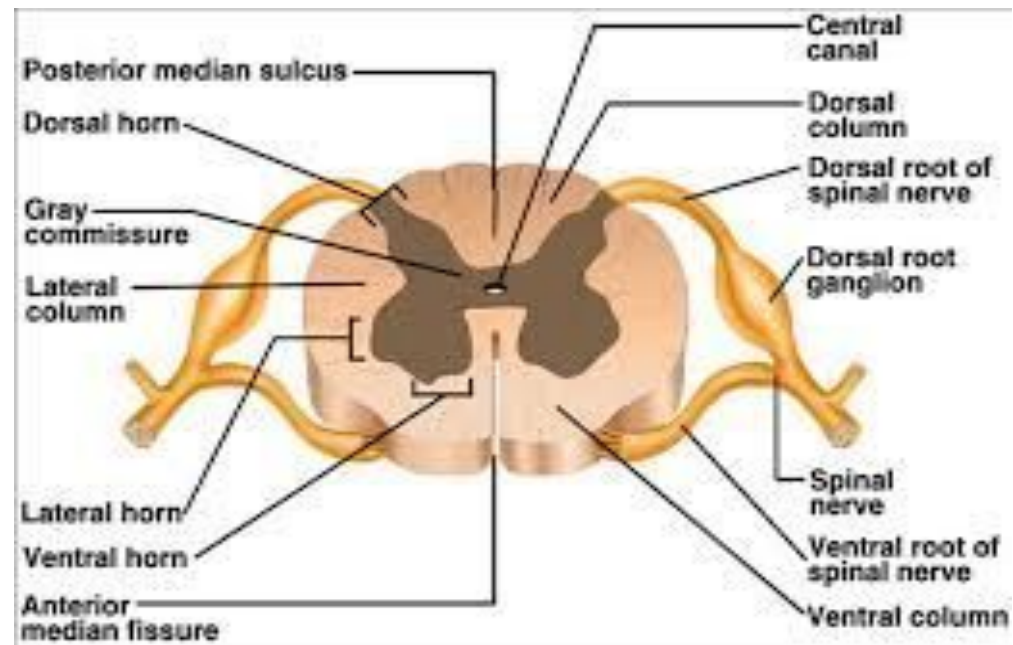
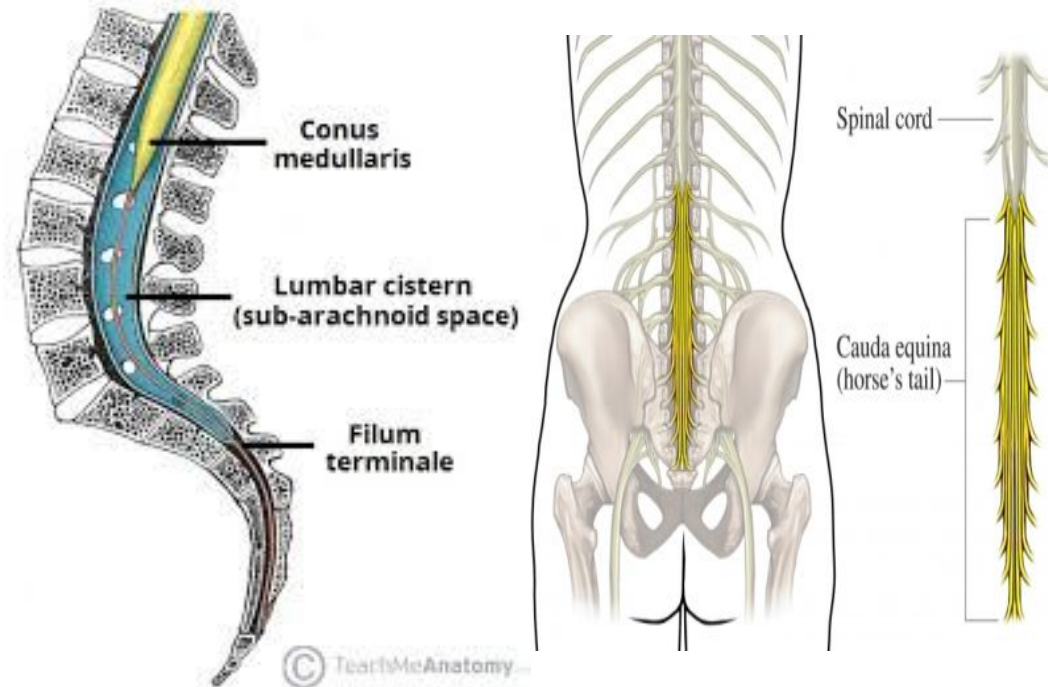
- **Describe external features of spinal cord.**
- **Explain the formation of a spinal nerve.**
- **Define the ascending tracts.**
- **Describe posterior white column and spinothalamic tracts and their functions.**
- **Define the descending tracts.**
- **Describe corticospinal tracts and their functions.**

Gross appearance:

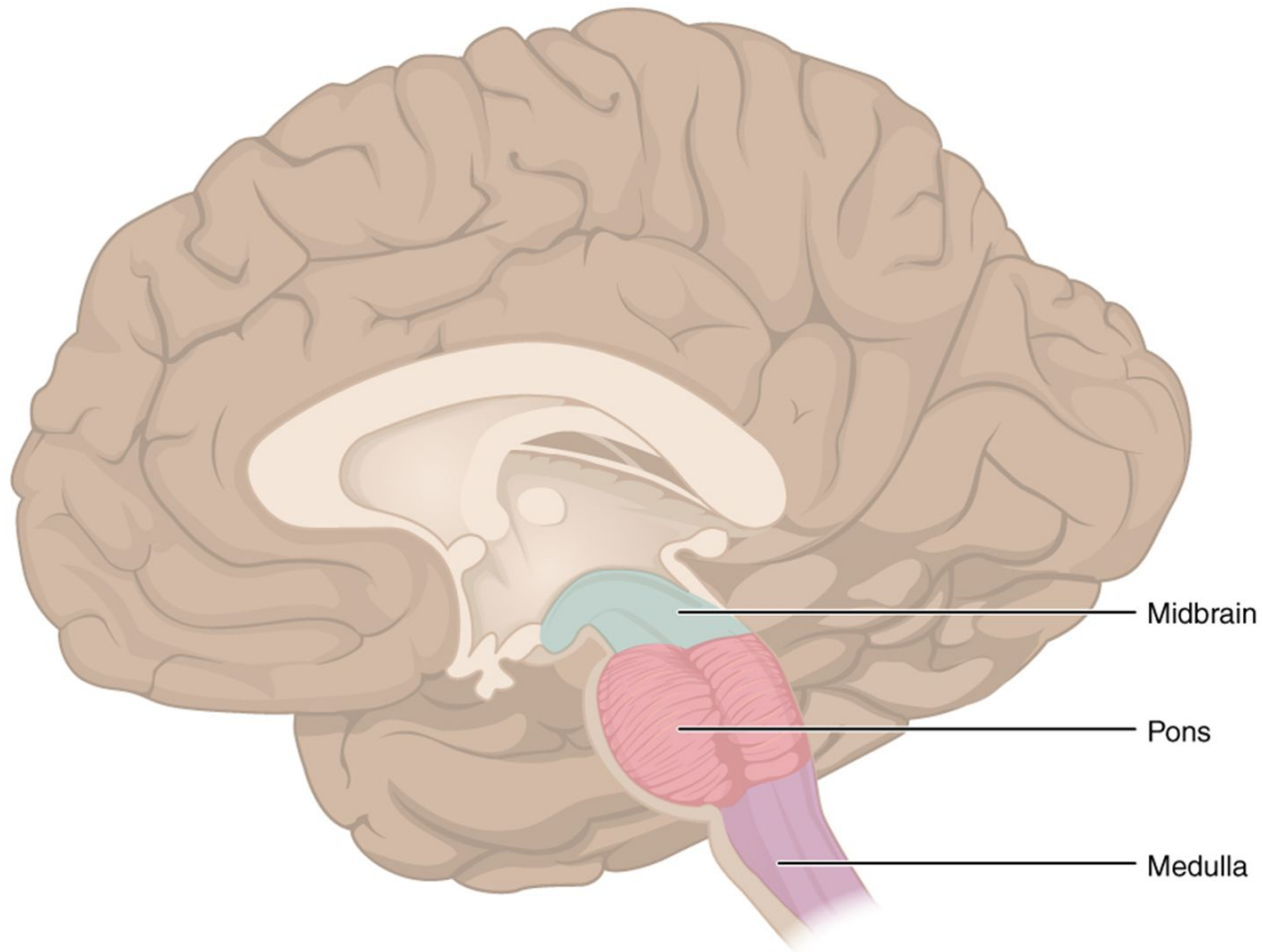
- **Begins** superiorly as at foramen magnum as a **continuation of medulla oblongata** of brain.
- In **adults**, **terminates inferiorly at lower border of first lumbar vertebra** and in **children** at **upper border of third lumbar vertebra**.
- Occupies upper two-thirds of vertebral canal and surrounded by meninges.
- Has fusiform cervical and lumbar enlargements.
- Inferiorly tapers off into **conus medullaris**.



- **Filum terminale** is a prolongation of pia mater from the apex of conus medullaris which descends and attach to the back of coccyx.
- **Cauda equina** is a bundle of spinal nerve roots that arise from the end of spinal cord below first lumbar vertebra in lower back.
- Has a deep longitudinal fissure anteriorly in the midline called **anterior median fissure** and a shallow furrow posteriorly called **posterior median sulcus**.

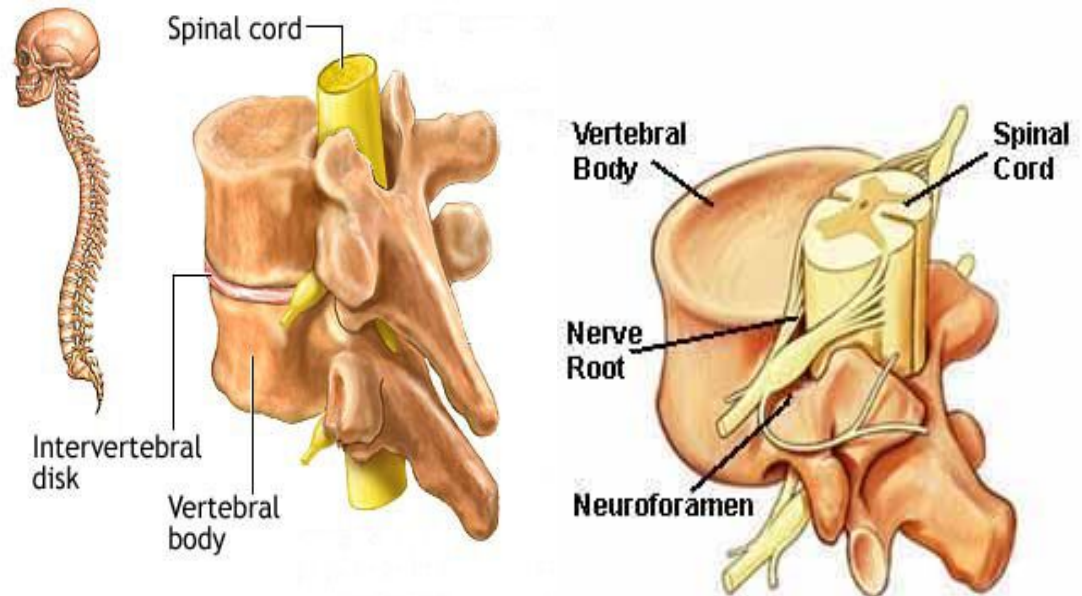
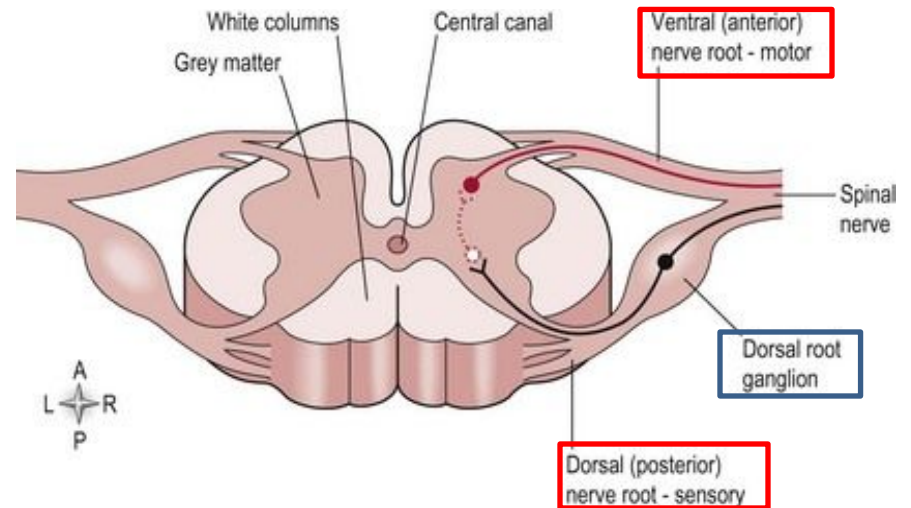


The Brainstem



How a spinal nerve is formed?

- The spinal cord gives off thirty one (31) pairs of spinal nerves.
- Each spinal nerve is attached to the spinal cord by an **anterior/motor root** and a **posterior/sensory root**.
- Each root has a series of rootlets which extend along the whole length of corresponding spinal segment.
- Each posterior nerve root has a **posterior /dorsal root ganglion**.

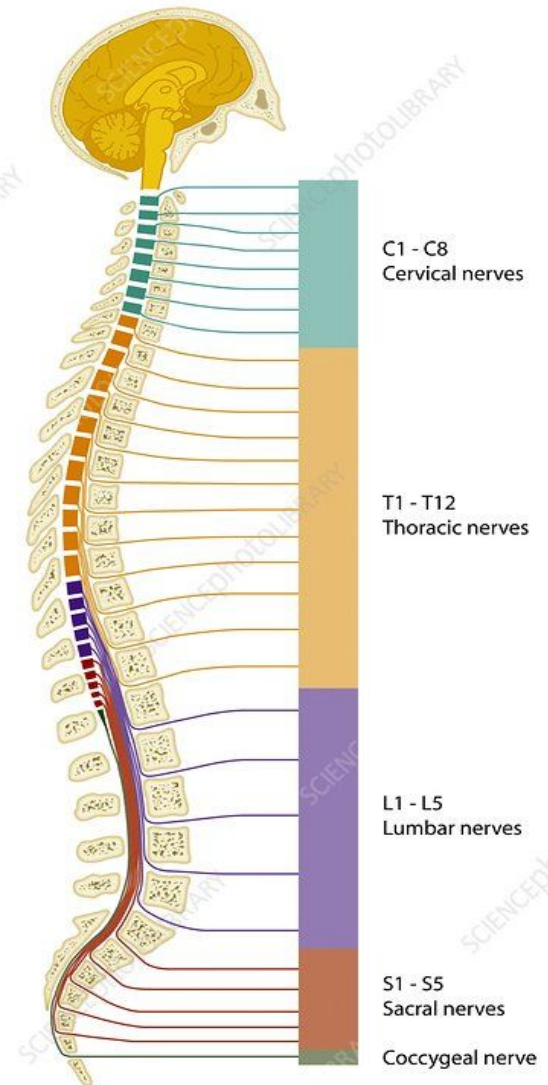
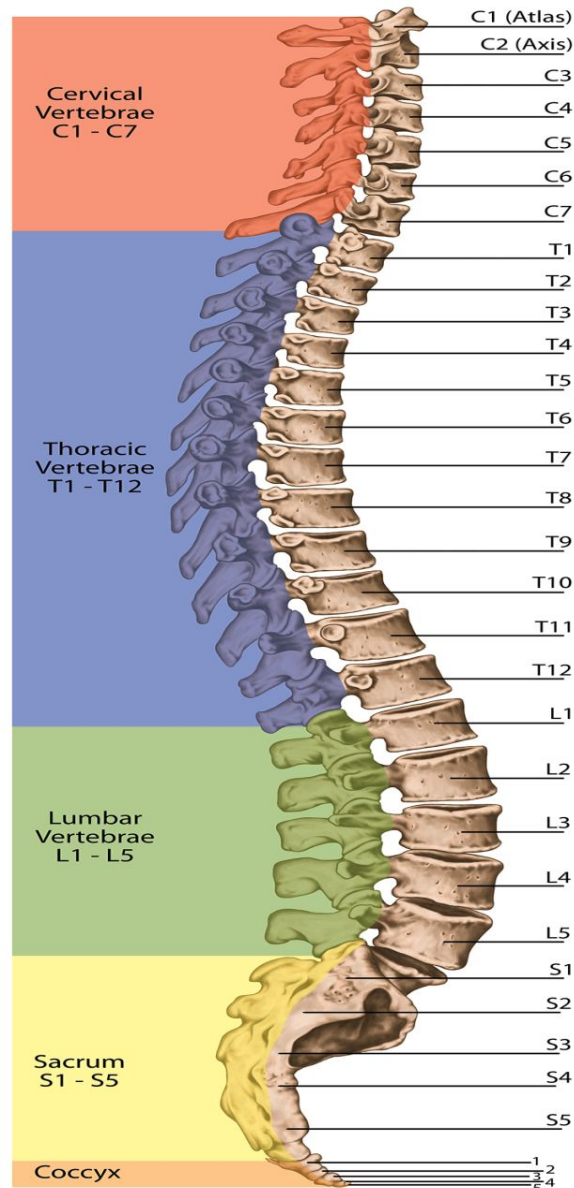
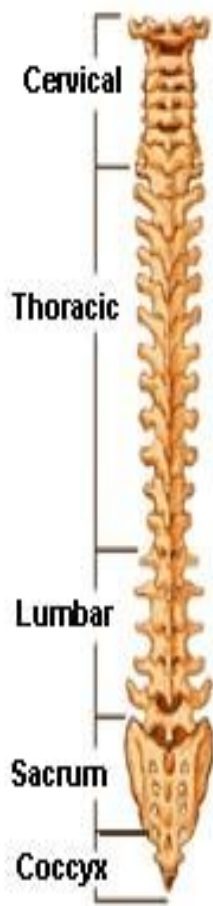


Regions of spinal cord:

Lateral (Side)
Spinal Column



Posterior (Back)
Spinal Column

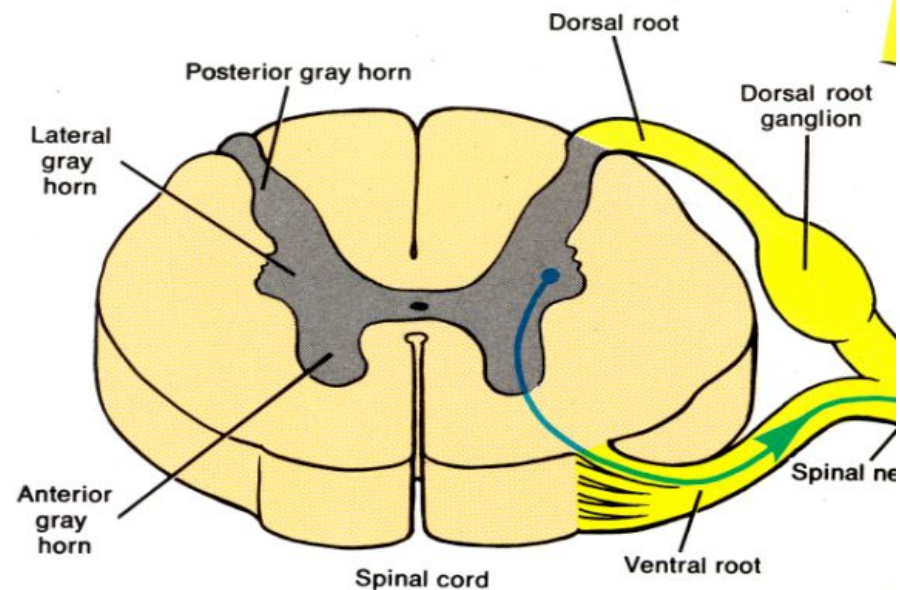
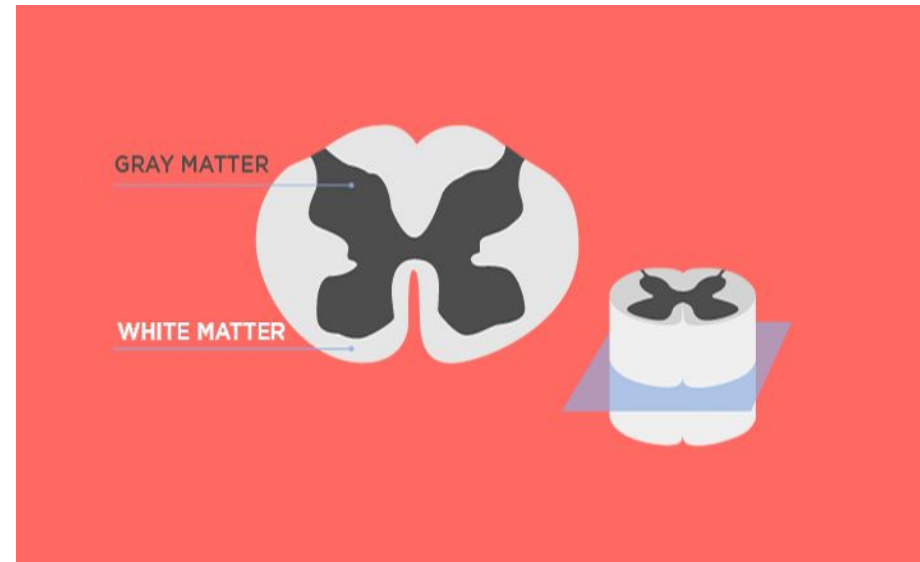


Internal structure:

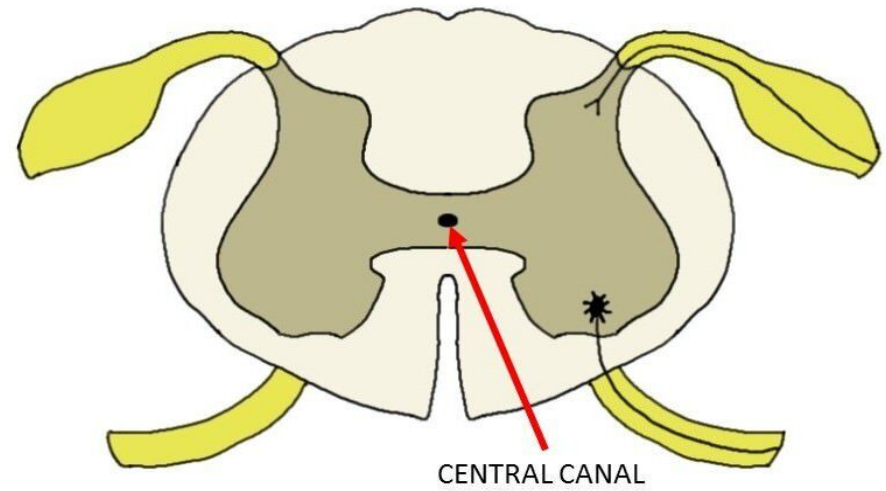
- The spinal cord has an **inner core of gray matter** surrounded by an **outer cover of white matter**.

THE GRAY MATTER:

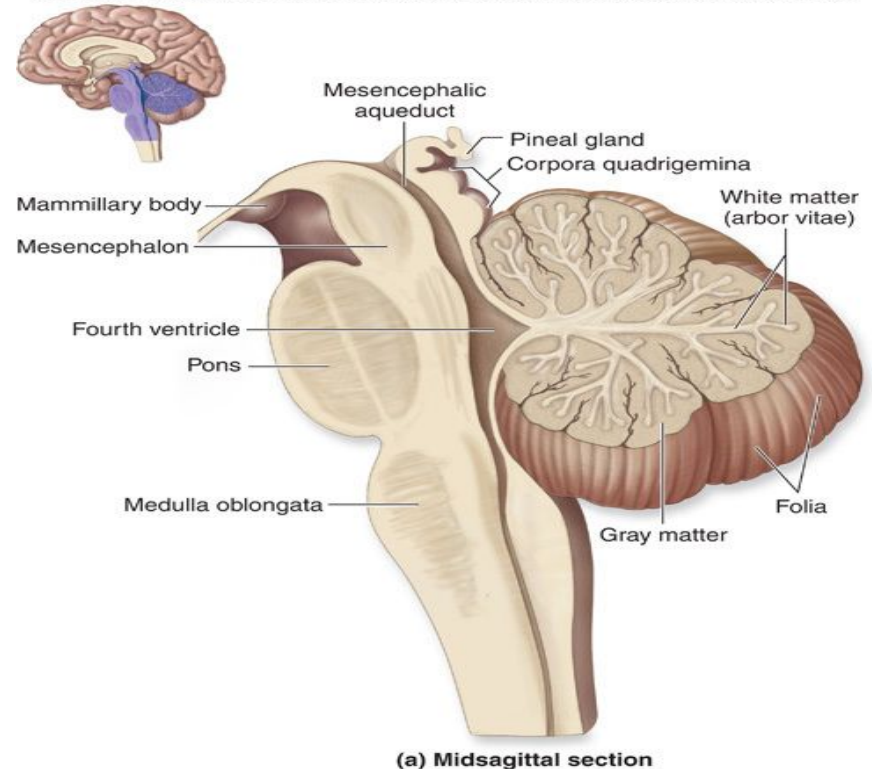
- Gray matter is **H-shaped** with **anterior and posterior gray columns** united by a thin **gray commissure**.
- A **lateral gray column** is present in thoracic and upper lumbar segments.
- A **central canal** is present in the center of gray commissure.



- Central canal opens superiorly in fourth ventricle through lower/caudal half of medulla oblongata and inferiorly terminates in filum terminale .
- It has a **fusiform dilatation** in conus medullaris called **terminal ventricle**.
- It is lined with ciliated columnar epithelium of ependyma and contains cerebrospinal fluid (CSF).

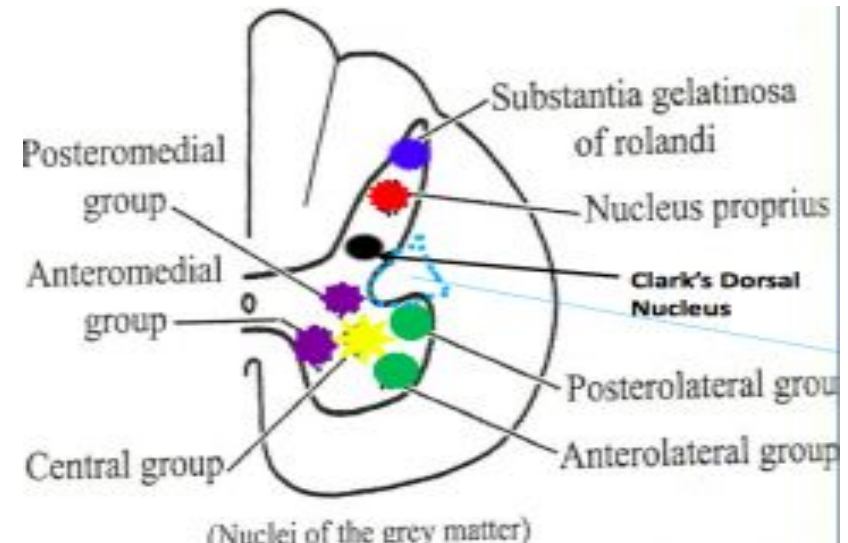
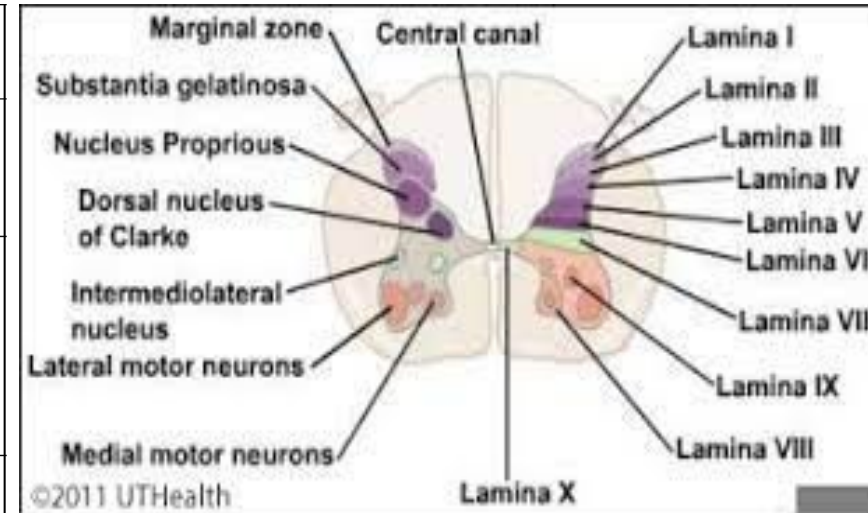


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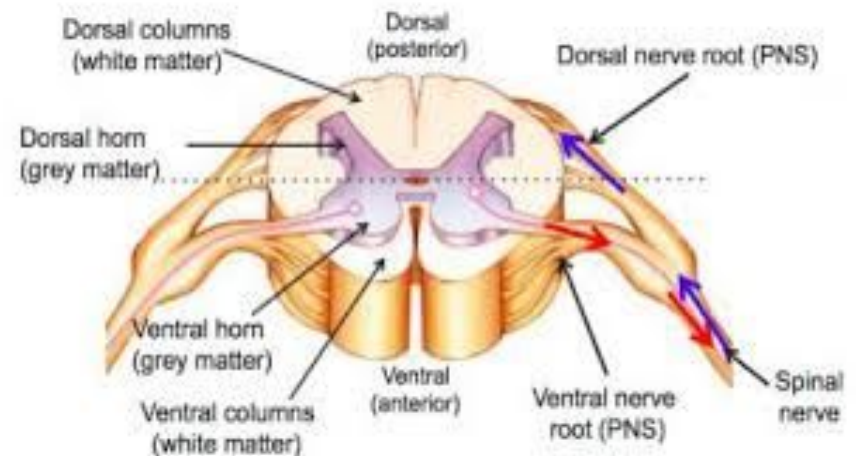
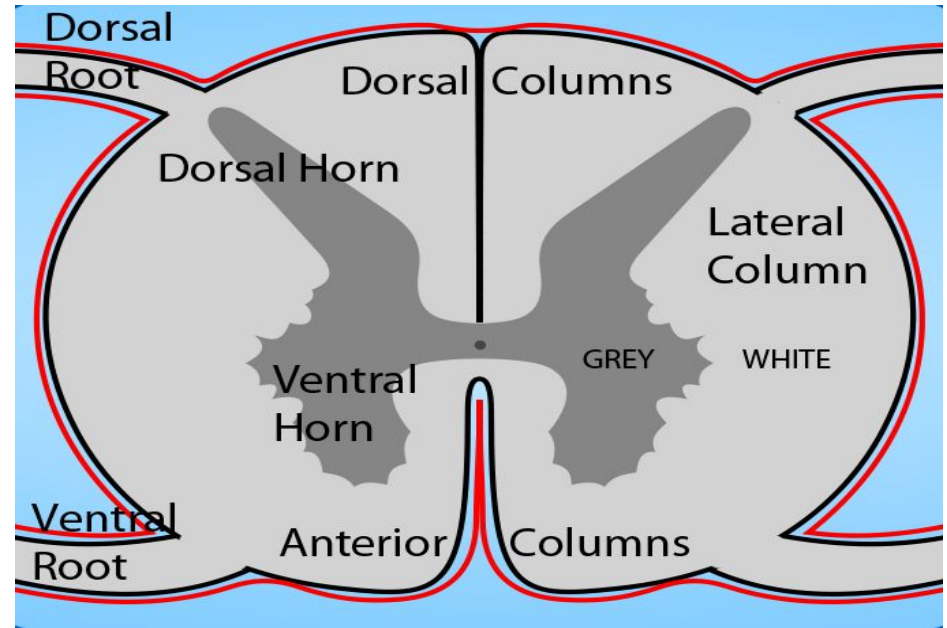
Nerve cell groups/nuclei in the GRAY MATTER of spinal cord

Region	Anterior Gray Column	Posterior Gray Column	Lateral Gray Column
Cervical	• Medial group of cells • Central group of cells • Lateral group of cells	• Substantia gelatinosa • Nucleus proprius	Absent
Thoracic	• Medial group of cells	• Substantia gelatinosa • Nucleus proprius • Nucleus dorsalis (Clarke's column) • Visceral afferent nucleus	Present
Lumbar	• Medial group of cells • Central group of cells	• Substantia gelatinosa • Nucleus proprius • Nucleus dorsalis (Clarke's column) • Visceral afferent nucleus	Present
Sacral	• Medial group of cells	• Substantia gelatinosa • Nucleus proprius	Absent



THE WHITE MATTER:

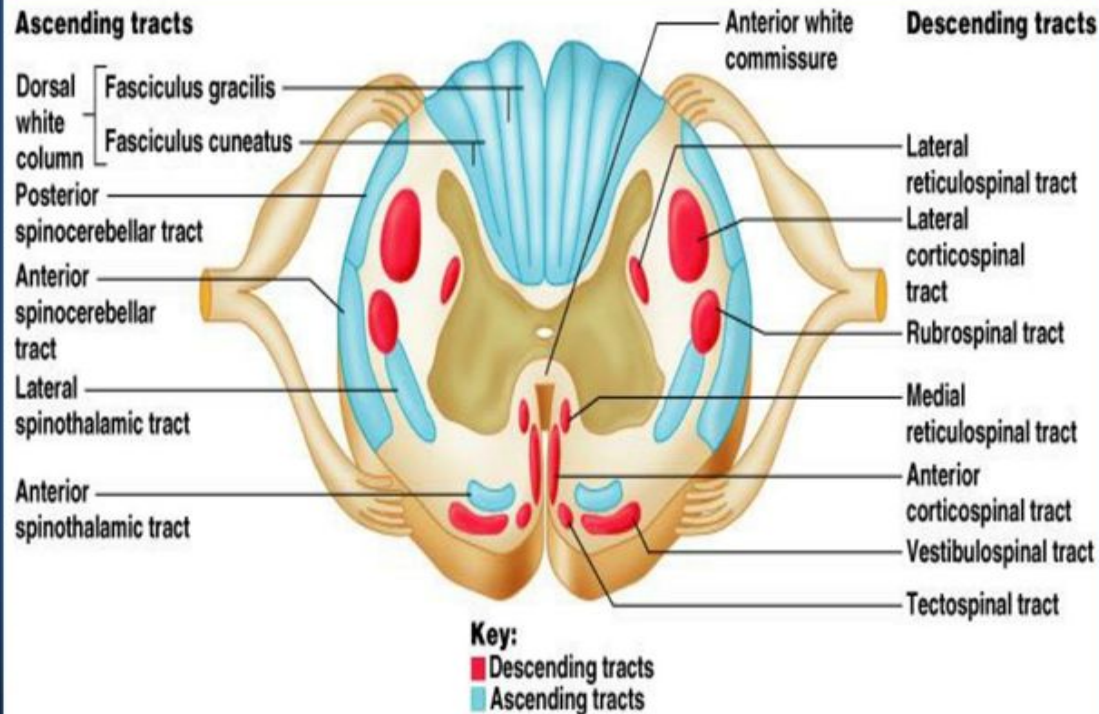
- White matter may be divided into anterior, lateral and posterior white columns/funiculi.
- **Anterior column** lies between midline and anterior nerve roots.
- **Lateral column** lies between anterior and posterior nerve roots.
- **Posterior column** lies between posterior nerve roots and midline.



The Ascending tracts:

- In the spinal cord sensory nerve fibers of different sizes and functions are segregated into bundles/tracts.
- Most of them ascend towards higher centers and thus connect spinal cord with brain.
- These bundles of sensory/ascending fibers are referred to as ascending tracts.
- The ascending tracts conduct afferent information which may be:
 - **exteroceptive** i.e. originates from outside the body such as pain, temperature and touch, *or*
 - **proprioceptive** i.e. originates from inside the body such as contraction of muscles, position and movements of joints etc.
- **Sensory receptors** are the peripheral nerve endings of afferent fibers which receive impulses.

Ascending Tracts of Spinal Cord



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ASCENDING TRACTS:

- LATERAL SPINOTHALAMIC TRACT
- ANTERIOR SPINOTHALAMIC TRACT
- FASCICULUS GRACIALIS
- FASCICULUS CUNEATUS
- DORSAL (post.) SPINOCEREBELLAR TRACT
- VENTRAL (ant.) SPINOCEREBELLAR TRACT
- SPINO-OLIVARY TRACT
- SPINOTECTAL TRACT

General schema of ascending pathway:

Generally ascending pathways consist of a three-neuron chain:

First-order neuron:

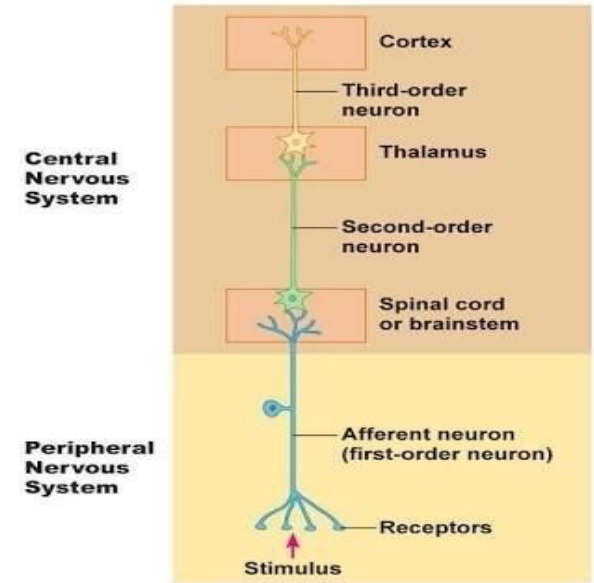
- Its cell-body lies in posterior root ganglion of spinal nerve.
- Its peripheral process connects with a sensory receptor ending, and
- Its central process enters spinal cord through posterior root to synapse on the second-order neuron.

Second-order neuron:

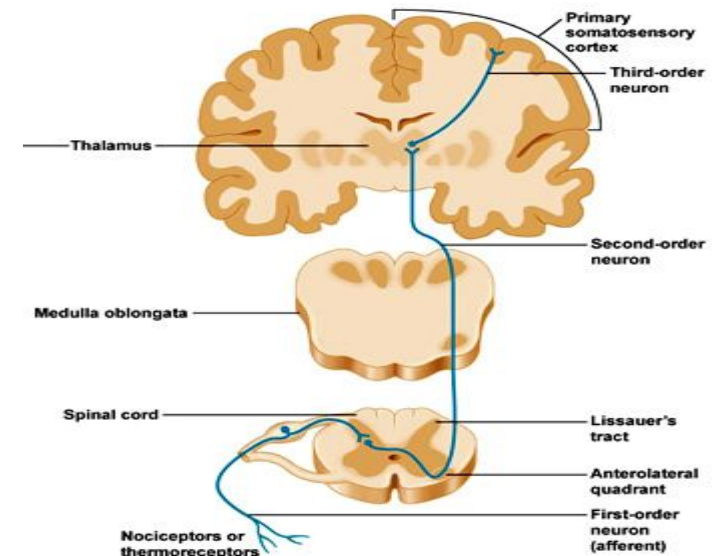
- Has an axon which crosses to opposite side (decussates) and ascends to a higher level of CNS to synapse with the third-order neuron.

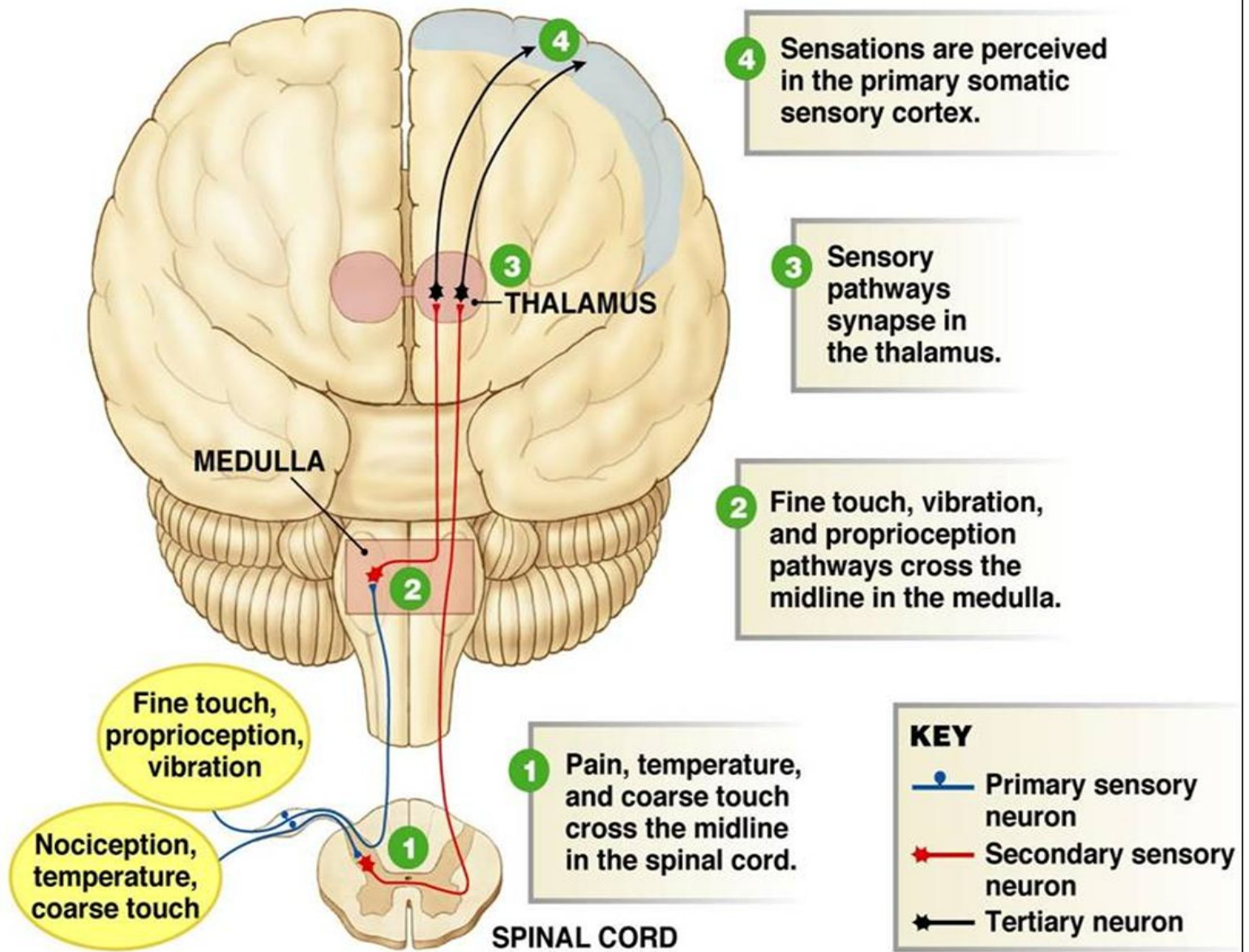
Third-order neuron:

- Its cell-body lies in thalamus which gives rise to a projection fiber that passes to a sensory area of cerebral cortex.



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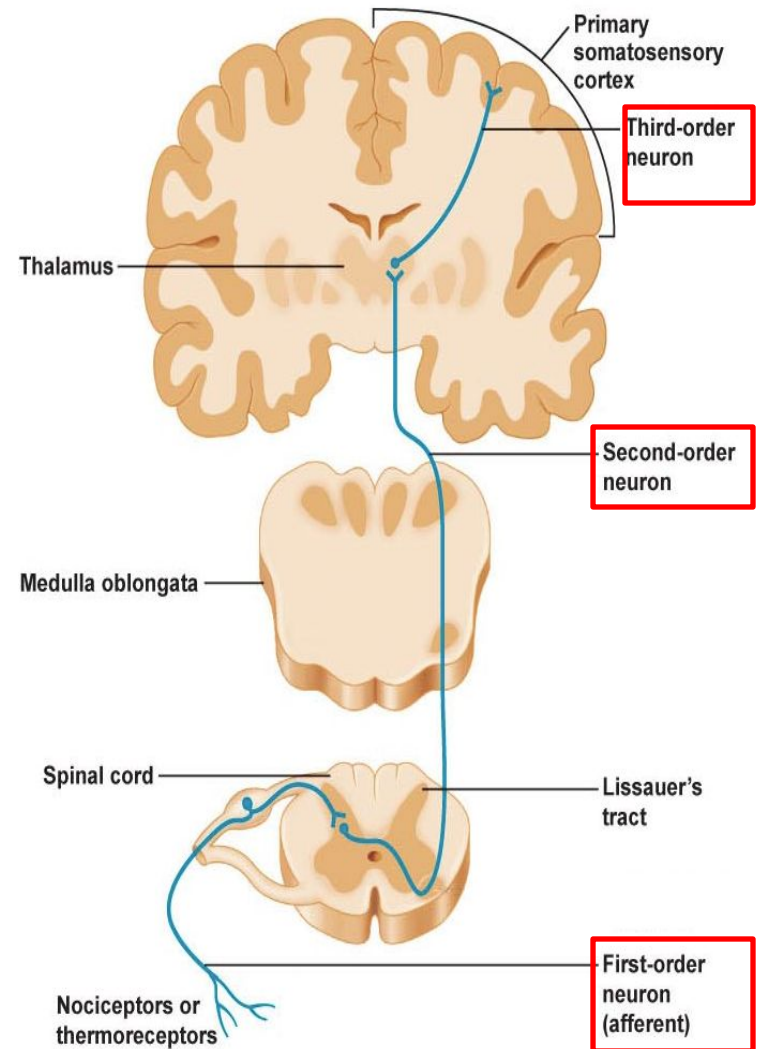




Spinothalamic tracts:

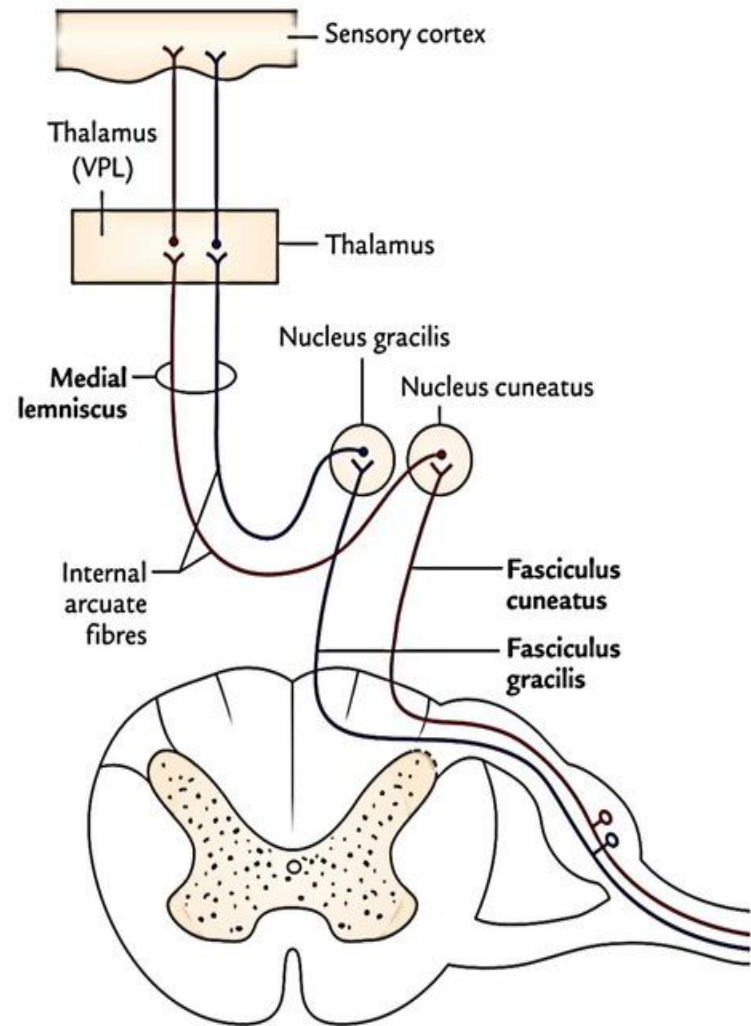
- **Lateral spinothalamic tract** carries **pain and thermal sensations**.
- **Anterior spinothalamic tract** carries **crude touch and pressure sensations**.
- Cell-body of 1st order neuron lies posterior root ganglion.
- 1st order neuron terminates by synapsing with 2nd order neuron in posterior gray column of spinal cord.
- Axon of 2nd order neuron now crosses obliquely (decussates) to other side of the cord and ascends in contralateral white column to synapse with 3rd order neuron in ventral posterolateral (VPL) nucleus of thalamus.
- Cell-body of 3rd order neuron in thalamus projects fibers through internal capsule and corona radiata to posterior central gyrus of cerebral cortex.

Spinothalamic Tract

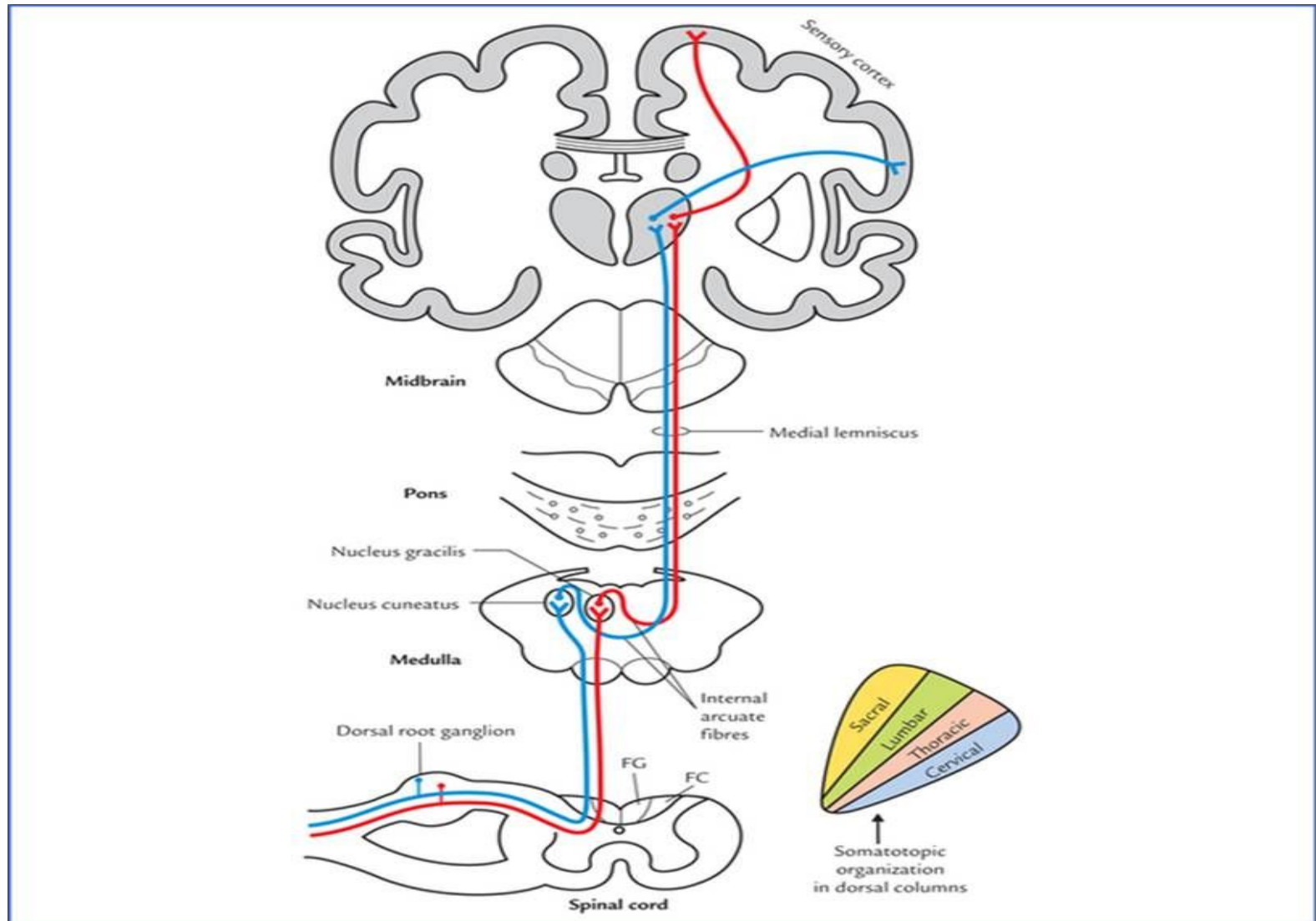


Posterior white column (Fasciculus gracilis and cuneatus):

- Carry sensations of **discriminative touch, two-point discrimination, vibratory sense and muscle joint sense**.
- Cell-body of 1st order neuron lies posterior root ganglion.
- 1st order neuron passes directly to posterior white column in spinal cord and ascends upward ipsilaterally as fasciculus gracilis and fasciculus cuneatus.
- 1st order neurons of fasciculus gracilis and cuneatus terminate by synapsing on 2nd order neurons in nucleus gracilis and nucleus cuneatus in medulla oblongata.
- The axons of 2nd order neurons called internal arcuate fibers decussate with the corresponding fibers of opposite side, in the sensory decussation and ascend as medial lemniscus to end by synapsing on 3rd order neurons in VPL nucleus of thalamus.
- 3rd order neurons of thalamus project fibers through corona radiata and internal capsule to post-central gyrus of cerebral cortex.



Schematic diagram of fasciculus gracilis & cuneatus:



Scheme of main ascending tracts/somatosensory pathways to consciousness

Pathways	Sensation	First-order neuron	Second-order neuron	Third-order neuron	Destination
Lateral spinothalamic	Pain and temperature	Dorsal/posterior root ganglion	Substantia gelatinosa in posterior gray column	Ventral posterolateral (VPL) nucleus of thalamus	Post-central gyrus
Anterior spinothalamic	Light/crude touch and pressure	Dorsal/posterior root ganglion	Substantia gelatinosa in posterior gray column	Ventral posterolateral (VPL) nucleus of thalamus	Post-central gyrus
Fasciculus gracilis and cuneatus	Discriminative touch, vibratory sense, conscious muscle joint sense	Dorsal/posterior root ganglion	Nucleus gracilis and nucleus cuneatus of medulla oblongata	Ventral posterolateral (VPL) nucleus of thalamus	Post-central gyrus

Spinocerebellar tracts ventral & dorsal

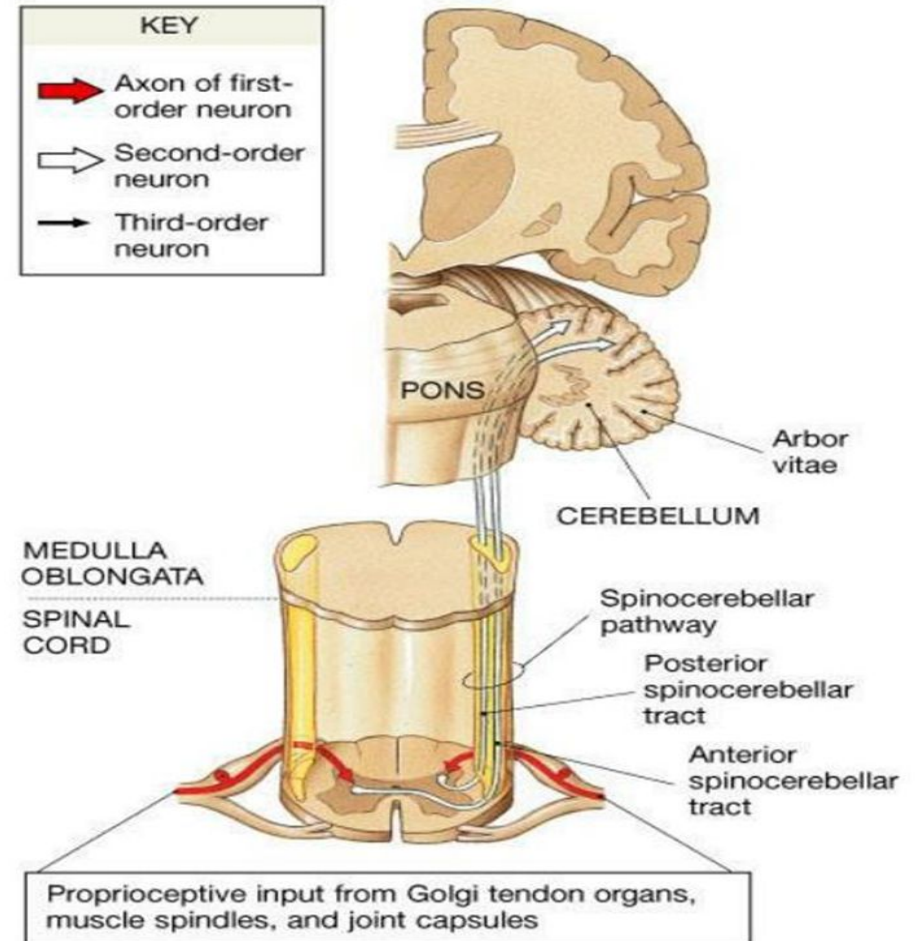
❖ Carry information from muscle spindle, Golgi tendon organ and tactile receptors to the cerebellum.

❖ To control posture and coordination of movement.

❖ The dorsal root ganglion is 1st order neuron.

❖ The 2nd order neuron cell bodies lie in the base of dorsal horn (Clark's).

❖ Terminate directly in the cerebellar cortex (vermis)

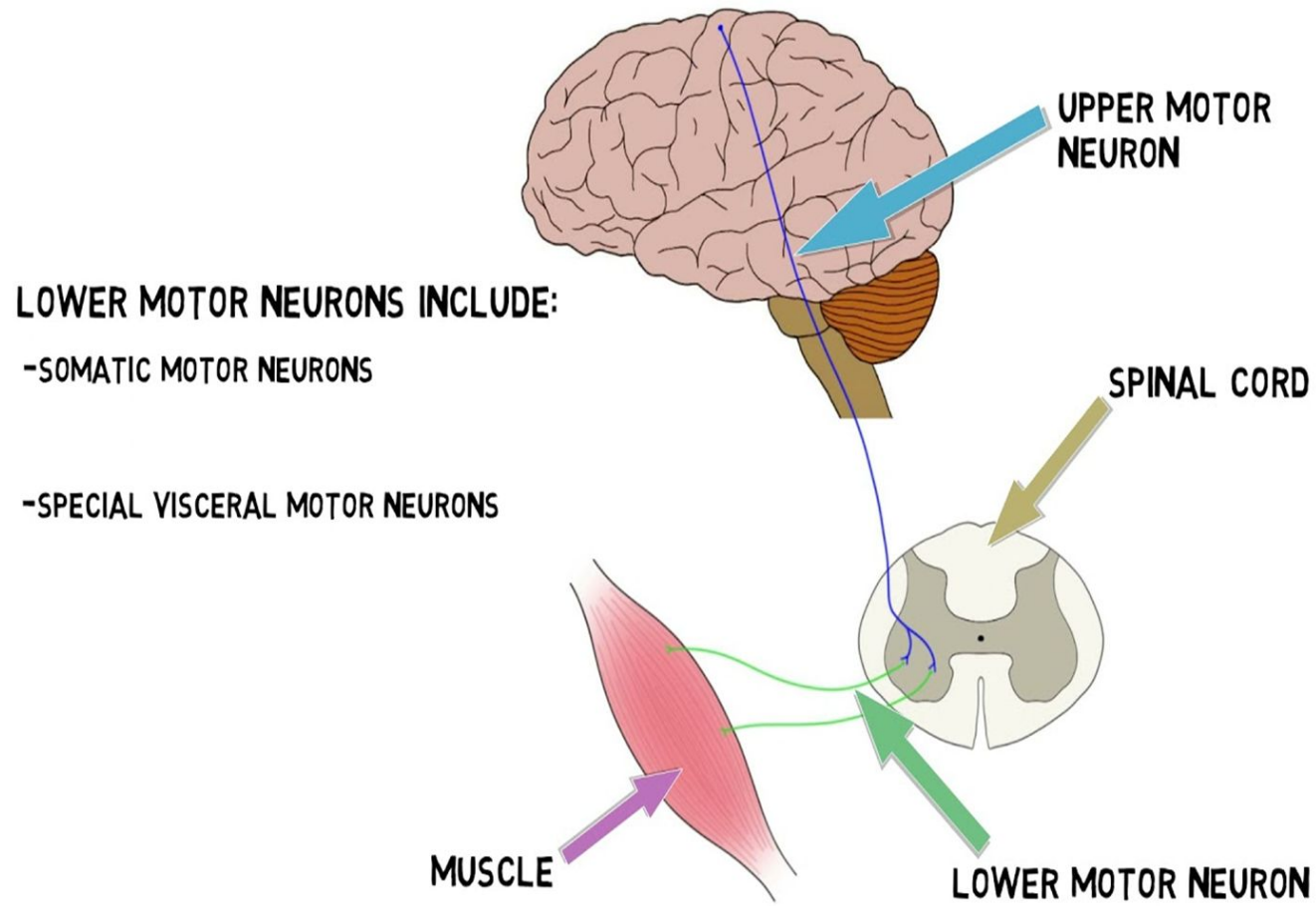


Muscle joint sense pathways to cerebellum				
Pathways	Sensation	First-order neuron	Second-order neuron	Destination
Anterior and posterior spinocerebellar	Unconscious muscle joint sense	Posterior root ganglion	Nucleus dorsalis	Cerebellar cortex

The Descending tracts:

- The descending tracts are the pathways by which efferent/motor signals are sent from brain to lower motor neurons.
- The lower motor neuron then directly innervate muscles to produce movements.
- The **lower motor neurons** are situated in the anterior gray horns of spinal cord and send axons to innervate the skeletal muscles through anterior spinal nerve roots.
- The **upper motor neurons** are the nerve fibers which originate from cerebral cortex/brainstem and their axons remain in CNS.

Scheme of upper and lower motor neurons:



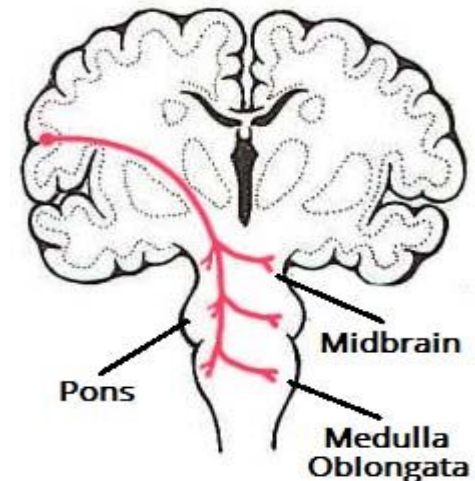
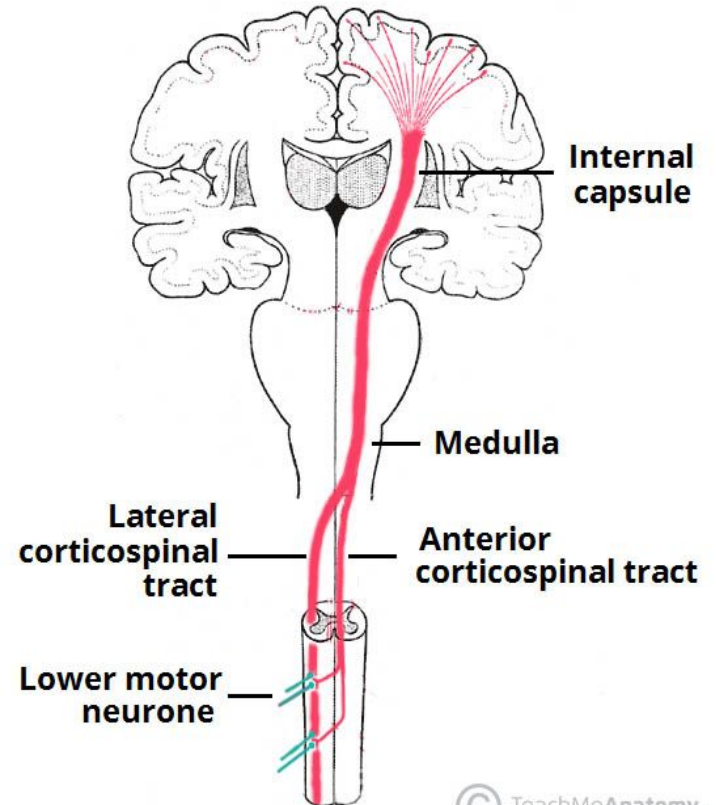
The descending tracts can be functionally divided into:

- **Pyramidal tracts** originate in the cerebral cortex, carrying motor fibers to the spinal cord and brain stem. They are responsible for the voluntary control of the musculature of the body and face.
- Example: **corticospinal tracts**.
- **Extrapyramidal tracts** originate in the brain stem, carrying motor fibers to the spinal cord. They are responsible for the involuntary and automatic control of all musculature, such as muscle tone, balance, posture and movement.
- Examples: reticulospinal tract, tectospinal tract, rubrospinal tract, vestibulospinal tract, olivospinal tract and descending autonomic fibers.

The Corticospinal tracts:

- Fibers of corticospinal tract arise as axons of pyramidal cells of (fifth layer of) cerebral cortex.
- Then the fibers converge in corona radiata and internal capsule, and
- Enter crus cerebri of midbrain, pons and then into medulla oblongata.
- In the most inferior (caudal) part of the medulla, the tract divides into:

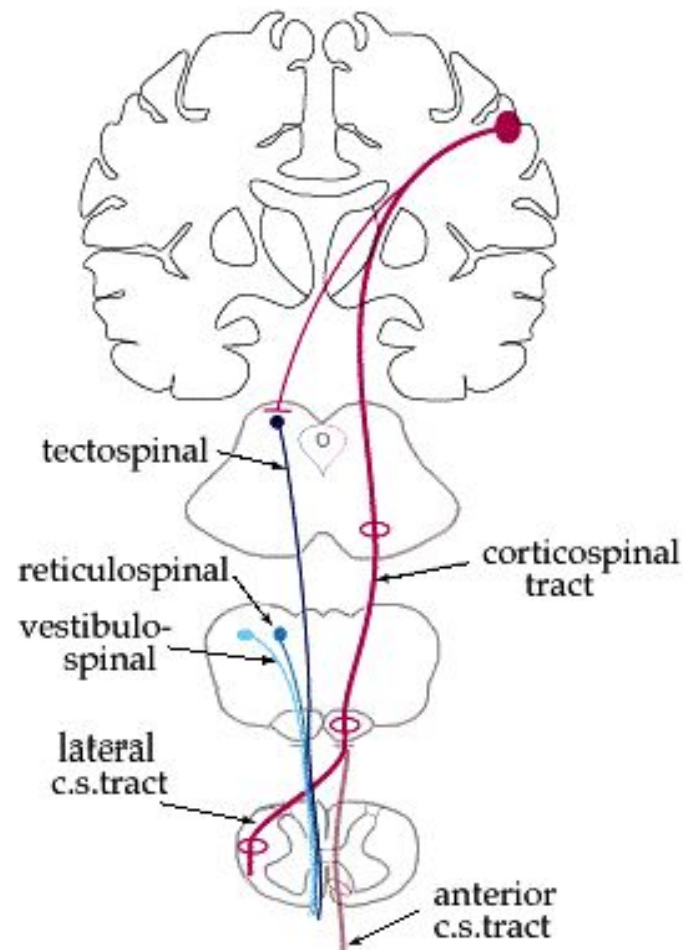
- **The lateral corticospinal tract:** the fibers cross midline at the junction of medulla and spinal cord, **decussation of pyramids**. They then descend into the spinal cord, terminating in the anterior/ventral gray horn. From the anterior gray horn, the lower motor neurons exit to supply the muscles of the body.
- The **anterior corticospinal tract** remains ipsilateral, descending into the spinal cord. They then decussate and terminate in the anterior gray horn of **cervical** and **upper thoracic** segmental levels.



Main Descending pathways to spinal cord

Pathway	Function	Origin	Cross/do not cross	Destination
Corticospinal tracts	Rapid, skilled, voluntary movements especially at distal ends of limbs	Primary motor cortex (pre-central gyrus)	Fibers of lateral corticospinal tract cross at decussation of pyramids whereas fibers of anterior corticospinal tract do not cross.	Anterior gray horn
Reticulospinal tracts	Inhibit/facilitate voluntary movement	Reticular formation	Some fibers cross	Anterior gray horn
Tectospinal tract	Reflex postural movements associated with sight	Superior colliculus	Cross soon after origin	Anterior gray horn
Rubrospinal tract	Facilitates activity of flexor and inhibits extensor muscles	Red nucleus	Cross immediately	Anterior gray horn
Vestibulospinal tract	Facilitates activity of extensor and inhibits flexor muscles	Vestibular nuclei	Do not cross	Anterior gray horn
Olivospinal tract	-----	Inferior olivary nuclei	Cross in brainstem	Anterior gray horn
Descending autonomic fibers	Control sympathetic and parasympathetic systems	Cerebral cortex, hypothalamus, amygdaloid nucleus, reticular formation		Sympathetic and parasympathetic outflows

Main Descending pathways to spinal cord



Injury to ascending tracts within the spinal cord:

- **Lateral spinothalamic tract injury:** contralateral loss of pain and thermal sensations below the level of lesion.
- **Anterior spinothalamic tract injury:** contralateral loss of light touch and pressure sensations below the level of lesion.
- **Fasciculus gracilis and cuneatus injury:** loss of muscle joint sense and position and movement of limbs of ipsilateral side below the level of lesion. Impaired muscular control, loss of vibration sense and tactile discrimination on the same side.

DIFFERENCE BETWEEN UPPER & LOWER MOTOR NEURON LESION

UMN LESION

- Paralysis affect movement rather than muscles
- Muscle wasting is only from disuse, therefore slight. Occasionally marked in chronic severe lesions.
- Spasticity of clasp-knife' type. Muscles hypertonic.

LMN LESION

- Individual muscle or group of muscles are affected.
- Wasting pronounced.
- Flaccidity. Muscles hypotonic.

Types of Paralysis:

- **Hemiplegia:** paralysis of one side of the body.
- **Monoplegia:** paralysis of one limb only.
- **Diplegia:** paralysis of two corresponding limbs i.e. arms or legs.
- **Paraplegia:** paralysis of two lower limbs.
- **Quadriplegia:** paralysis of all four limbs.

References:

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Richard S. Snell
8th edition
Chapter 4
Pages # 137-165

- Prepared by:
DR. MARIA KHAN
ASSISTANT PROFESSOR
OF ANATOMY
(DIKIOHS-DUHS)